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The Heretical Physicist: Revisiting Einstein's Objection to the Event Horizon

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This paper revives Einstein's original objection to the event horizon by arguing that horizon crossing is not physically realizable within finite proper time. Using a thought experiment involving an inelastic rope extended from a stationary observer to a falling body, we show that the proper rope length required diverges before the body reaches the horizon, yet the observer measures the rope as finite and sees its descent slow, like cars approaching a traffic jam. We argue this is a literal description of physical events, not an illusion, based on information available to the observer. This suggests the horizon is not a surface through which matter can pass, but a null boundary approached asymptotically. We critique the use of coordinate transformations that remove the Schwarzschild singularity, arguing they obscure the distinction between space and time. We further examine gravitational collapse, concluding that a massive body forms a “frozen” configuration rather than a singularity. By restoring operational continuity to causal structure near the Schwarzschild radius, we challenge the legitimacy of the event horizon as a physically traversable surface. If correct, this analysis may also impact our understanding of black hole formation and cosmic singularities such as the big bang.

Keywords: Event horizon; gravitational collapse; Schwarzschild metric; frozen star; causal structure

1. Introduction

Black holes are supposed to be one-way doors. Anything passing the event horizon is thought to be forever cut off from the rest of the universe, doomed to spiral inward toward a singularity. This picture has become orthodoxy, bolstered by decades of increasingly sophisticated theoretical models and indirect observational evidence. But like many ideas in physics, its ubiquity has numbed us to the original doubts. This paper revisits one of those doubts: the one raised by Einstein himself.

Einstein objected to the idea that a horizon could be physically traversed. The horizon is a null surface, traced out by outward-moving light rays that

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never escape. Nothing, not even light, can cross from inside to outside once it reaches this boundary. This is not due to a physical obstruction, but because, in the external frame, time dilation becomes so extreme that all motion, whether inward or outward, appears frozen. If we take this seriously, we must recognize that even light emitted from just above the horizon never escapes in any finite coordinate time. More provocatively, we must consider the possibility that no object ever actually crosses it at all.^a

Einstein's 1939 model involved a configuration of pressureless dust in circular orbits, which has since been shown to be unstable. However, the deeper insight of that paper was not the model itself, but the philosophical objection it implied: that spacetime near the Schwarzschild radius does not admit a smooth, continuous physical evolution. It is this core intuition, not the outdated specifics, that we revisit here using updated methods and causal reasoning.¹

This is not a new idea. The frozen star model, originally proposed by Oppenheimer and Snyder,² held that collapse stalls near the Schwarzschild radius when viewed from a distant frame. Later interpretations rebranded this as an illusion, insisting that infall continues smoothly in proper time. Coordinate transformations were invoked to justify this interpretation, effectively removing the singularity at the Schwarzschild radius and making room for interior evolution.^{3; 4} But this move may have been a framing misdirection.

In this paper, we argue that such transformations fail to preserve the local operational structure of spacetime. We introduce a thought experiment involving an inelastic rope and a stationary observer to demonstrate that horizon crossing is not only suspect in coordinate time, but also in proper time. We then examine the causal implications of this finding and explore an alternative interpretation: that the black hole is a frozen object, never completed, whose horizon is never crossed.

2. Einstein's Objection

Einstein remained unconvinced by the black hole interpretation of the Schwarzschild solution. In correspondence and interviews, he expressed skep-

^aClaims about the size or growth of a black hole, including its event horizon, are necessarily made from an external frame. Yet such measurements presume that infalling mass-energy has already crossed the horizon. If no signal or causal process can confirm this from the outside, a strict operationalist would question whether such a claim is physically meaningful.

ticism about the idea of matter vanishing into singularities and was wary of attributing physical meaning to mathematical features that defied operational verification.

In 1939, Einstein published a paper in *Annals of Mathematics* arguing that Schwarzschild singularities do not occur in reality.¹ His approach was clunky by modern standards: he modeled a spherically symmetric cloud of particles orbiting at relativistic speeds and concluded that gravitational collapse halts before forming a horizon. Although this particular construction has been superseded, the underlying intuition remains valuable.

Einstein's hesitation centered on causality. A horizon, by definition, marks a boundary beyond which no information can reach an outside observer. But if we cannot observe or interact with events beyond the horizon, then what justifies treating them as physically real? Should we not, as operationalists, refrain from asserting the existence of entities or regions that no possible measurement can ever confirm?

The modern rebuttal to Einstein invokes proper time: while a distant observer sees infall freeze at the horizon, the infalling body itself supposedly crosses in finite time. But this distinction depends on a global coordinate transformation that mixes time and space in a way that may not preserve local causal structure.^{5; 6} We argue that this sleight of hand cannot be relied upon to resolve the tension.

3. No Rope Long Enough



To make this scenario concrete, imagine that a clock is attached to a massless, perfectly inelastic rope. A stationary observer located far from the Schwarzschild radius unreels this rope, allowing the clock to fall freely. The rope unreels without tension, simply following the clock's motion downward. At each instant, the observer measures the proper length of rope required to maintain contact with the clock, which corresponds to the spatial distance along a constant-time hypersurface from her location to the clock's radial position.

Fig. 1. An observer unreels an inelastic rope while tracking light pulses from a falling clock.

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^b

Let the clock emit pulses of light, but now with a twist: it adjusts both the rate and wavelength of each pulse as it falls, in order to compensate for gravitational redshift. Specifically, if the observer is to receive one pulse per second of her proper time, the clock must flash at a rate

$$f(r) = \frac{1}{\sqrt{1 - \frac{2M}{r}}}, \quad (3.1)$$

which increases as the clock descends. Similarly, to ensure that each pulse arrives with a fixed observed wavelength λ_0 , the clock must blueshift its emitted wavelength to

$$\lambda(r) = \lambda_0 \sqrt{1 - \frac{2M}{r}}. \quad (3.2)$$

These adjustments perfectly cancel the effects of gravitational redshift.⁶ As a result, the observer receives a steady stream of pulses—each one second apart, each with identical wavelength—indefinitely. There is no fading, no slowing, and no final flash. The clock never disappears; it remains in perfect communicative contact.

The spatial distance required to keep the rope taut is given by the integral

$$\ell = \int_{2M+\epsilon}^{r_0} \left(1 - \frac{2M}{r}\right)^{-1/2} dr, \quad (3.3)$$

which diverges logarithmically as $r \rightarrow 2M$.⁵ This means the rope must become infinitely long to reach the horizon. Even though the falling mass approaches the horizon in finite proper time, the required rope length grows without bound in the external frame. This is not a practical failure; it is a geometric impossibility. The rope extends farther and farther, but never reaches the surface.

^bThe perfect inelasticity of the rope is an idealization, necessary only if the observer intends to retrieve the clock at arbitrarily late times. In a realistic rope, any attempt to stop the fall would propagate at the speed of sound, which—near the horizon—would eventually be too slow to reach the descending clock. But this limitation is irrelevant. The key point is that the observer continues receiving pulses indefinitely and can verify at any time that the rope remains finite in length. That alone suffices to conclude that, from her perspective, the clock has never crossed the event horizon.

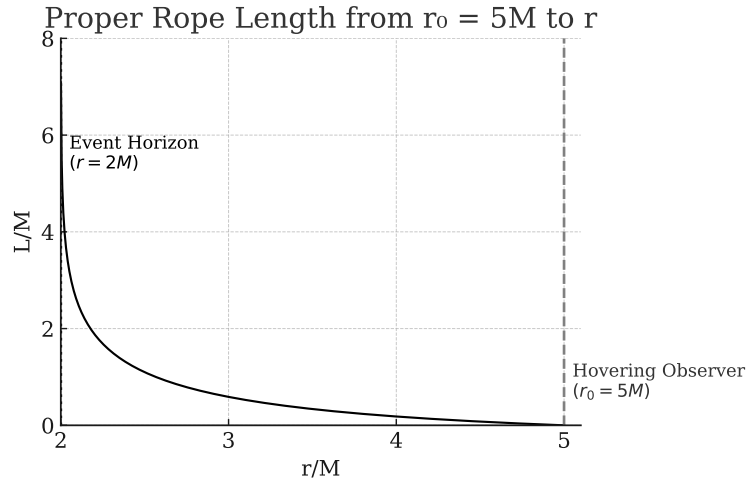


Fig. 2. Proper rope length diverges as the clock nears the event horizon.

This is more than a logistical inconvenience. At any moment—now or in the distant future—the observer may stop the descent, verify that the rope remains finite in length, and confirm that the clock is still flashing. Because the clock flashes at a rate precisely tuned to deliver one pulse per second, and each pulse arrives with a constant wavelength, the observer continues to receive them without interruption. She may even retrieve the clock by pulling it back up. The idea that the clock has already crossed the event horizon, ever, is therefore untenable. For any finite time, the clock remains outside, observable, and mechanically recoverable.

This leads to a profound conclusion: the fall is never completed in any operational sense. Although the test mass experiences finite proper time, the observer witnesses only an asymptotic approach. There is no measurable event that marks the crossing, and no mechanism to confirm that it has occurred. The physical content of the fall ends not at the horizon, but just before it.

This reframes the causal structure of collapse. We are accustomed to imagining the sequence of events as dictated by proper time—first the fall, then the crossing. But in this case, the continuous interactions between the mass and the observer, mediated by light pulses and mechanical tension, persist while the mass remains outside the horizon. The rope can be pulled back. The clock can be recovered. There is no subsequent event to wait for, no transition to confirm. The notion of crossing is not merely unobserved; it

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is causally excluded. The horizon is never reached—not in principle, not in theory, and not in reality.

4. The Event Horizon as a Zero-Way Surface

The rope experiment forces a reevaluation of the event horizon’s physical role. Traditionally imagined as a one-way surface, admitting infall but forbidding escape, the horizon is shown here to be more restrictive. It does not merely delay signals or distort observations; it halts motion altogether. Despite releasing an unlimited amount of rope, the observer never loses causal contact. No part of the system is ever beyond reach. This is not a one-way relationship. It is a failure of passage in both directions. The rope does not snap. The clock does not vanish. The light pulses never cease. Every signal, every force, and every verification remains accessible. In this sense, the horizon behaves not as a membrane to be crossed, but as a causal terminus. We call it a zero-way surface.

Crucially, this conclusion arises from a system in perfect mechanical continuity. The observer, the rope, and the clock begin as a single, unified apparatus. If the horizon could be crossed, then some event must sever that continuity. Yet none is observed. There is no moment when the clock becomes unreachable, no evidence of a causal boundary being breached. The geometry itself forbids completion of the fall. This reframes the question of gravitational collapse. Matter does not pass through the horizon into an interior region. Instead, collapse asymptotically approaches the boundary and halts. The result is not a singularity, but a stable, compact object whose surface lies infinitesimally above the Schwarzschild radius, a configuration often called a frozen star.⁸

The key insight is not that the clock fails to cross, but that nothing requires it to. General relativity, when interpreted operationally rather than globally, does not demand that horizons form in finite time. It only appears to do so when one adopts coordinates that extend across causally disconnected regions. But from the standpoint of physical interaction, the crossing never happens, and that fact is measurable, not metaphysical.

5. Frozen Collapse

If the event horizon functions as a zero-way surface, one that cannot be crossed even in principle, then the standard picture of black hole formation must be revised. Collapse does not continue unimpeded past a horizon; instead, it stalls asymptotically just above it. The endpoint is not a singularity

shrouded by a horizon, but a compact object whose surface remains forever visible in principle, though increasingly obscured by redshift. This is the frozen star.^{2; 8}

A frozen star forms when infalling matter compresses beyond all known forms of degeneracy pressure. Its surface asymptotically approaches the Schwarzschild radius but never crosses it. From an external perspective, it becomes virtually indistinguishable from a black hole: no light escapes detectably, and the gravitational field mimics that of a true event horizon. Yet the key distinction remains—there is no interior region causally severed from the outside.

This outcome does not require quantum corrections or exotic physics. It emerges naturally from classical general relativity when causal structure is taken seriously.^{5; 6} Near a null surface, radial distances stretch and signal propagation slows indefinitely. Infalling matter, viewed operationally, never completes the journey. No singularity forms. No horizon is crossed.

The frozen star also sidesteps long-standing paradoxes associated with traditional black holes. Information is not lost, because it never passes into an unreachable interior. No firewall is needed to enforce unitarity, because the horizon, strictly speaking, never forms. Unlike many proposed resolutions that rely on quantum gravity or modified collapse models, the frozen star arises without modifying the Einstein equations. It simply interprets their causal structure as physically binding.

Recent studies of evaporating black holes in dynamical spacetimes suggest that the standard assumptions behind horizon formation may not hold in realistic collapse scenarios.^{9; 10} The frozen star model supports this trend, not by altering the theory, but by clarifying what it already implies when coordinate extensions are set aside and causal access is treated as the defining feature of physical existence. A similar interpretation appears in Hahne's gradual-construction model, in which the black hole forms externally, with gravity intensifying incrementally rather than all at once.

Note: while the Schwarzschild radius is typically treated as a fixed threshold for horizon formation, this presumes the existence of a fully formed event horizon in the first place—exactly the point under dispute. The literature is notably sparse on how, exactly, such a horizon emerges during collapse. Intuition from classical mechanics suggests that gravitational collapse would begin at the center and proceed outward, not occur simultaneously at all radii. In such a scenario, any horizon-like structure would originate near $r_0 = 0$ and expand. But the geometric limits on causal contact apply equally

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to the surrounding mass during this process. The result is the same: crossing remains unachievable, regardless of how the horizon attempts to form.

This raises a natural question: if collapse halts at the center, where does the matter go? In the frozen star model, the material content of the collapsing object is compressed not into a central singularity, but into an increasingly dense shell near the surface. From the outside, the gravitational field is indistinguishable from that of a black hole, but the internal structure is radically different. Rather than accumulating at a central point, the mass effectively “backs up” near the boundary due to the formation of a null surface at the core. This does not imply a physical force pushing matter outward, but a causal bottleneck that prevents further infall. The interior becomes inert—not singular, but unreachable. No signal or particle completes its journey to the center in finite time. The collapse thus inverts: what appears to be contraction toward a singularity is, causally, a stalling of evolution from the inside out. This structure may seem strange, but it is the natural consequence of interpreting the Einstein equations as constraints on causal continuity rather than geometric extension.⁷

6. Implications for Black Holes and the Big Bang

If the event horizon is not a physical gateway but a causal boundary that cannot be crossed in finite time, then the traditional picture of singularities—whether buried inside black holes or posited at the origin of the universe—requires serious reevaluation. These constructs rely on a breakdown of causal structure and a willingness to stitch together incompatible patches of spacetime, often using mathematical tricks to bury physical infinities behind unreachable surfaces. The findings of this paper offer an opportunity to restore geodesic completeness as a guiding principle of physics: no region of spacetime should permit a violation of causal order or conceal divergent values. In this view, the big bang may not represent a true origin point, but rather an asymptotic causal boundary approached by all backward-directed trajectories. Similarly, gravitational collapse halts not in a singularity but at a boundary of causal inaccessibility. This framework preserves operational continuity and eliminates the need to invoke physically untestable infinities. A more detailed analysis of these broader implications will be presented in a future work.

7. Renormalization and the Fungibility Fallacy

The argument that black holes are smooth and continuous objects, with a traversable horizon and an interior geometry, relies heavily on coordinate transformations that remove apparent singularities. Kruskal–Szekeres coordinates, for instance, recast the Schwarzschild metric into a form that is analytic at $r = 2M$, inviting the interpretation that the horizon is not a physical boundary but a removable coordinate artifact.^{3; 4}

But this move hides a deeper problem. Coordinate transformations are not physically free. They must preserve causal structure to preserve physical meaning. When a transformation converts a null surface into a regular one, or interchanges the roles of time and space, it does more than change perspective—it alters the operational character of spacetime.⁵

This becomes obvious when we consider an analogy. Imagine a mountain, stationary and massive. Now perform a coordinate transformation that redefines time and space such that the mountain appears to move at light speed and has vanishing rest mass. Mathematically, this is permissible. Physically, it is nonsense. We cannot transform away mass, motion, or causality just by relabeling coordinates.

Yet this is precisely what is done with the Schwarzschild horizon. A divergent time dilation is recast as a smooth passage. A surface from which no information returns is promoted to a region that still evolves and interacts. The transformation swaps the inaccessible for the accessible, and claims physical continuity. But the continuity is semantic, not causal.

This is what we call the *fungibility fallacy*—the mistaken belief that time and space are interchangeable commodities under coordinate transformations. In reality, they play fundamentally different roles. Time orders causality. Space permits simultaneity. Swapping them does not merely relabel a diagram—it reassigns what events can influence one another.⁶

Any coordinate system that permits an observer to cross the event horizon must therefore be treated with suspicion. If the transformation requires a redefinition of time that breaks contact with prior causal structure, then it does not describe a physical process. It describes an imagined world, stitched together from incompatible frames.

We must resist the temptation to treat all coordinate systems as physically equivalent. Mathematics permits such transformations. Physics may not.

8. Conclusion: Causality Restored

Einstein’s objection to the event horizon was not rooted in a misunderstanding of general relativity, but in a clear intuition about the nature of physical reality. He understood that causality—not coordinate freedom—must govern our interpretation of spacetime. Horizons, if they are to be taken seriously, must be defined by their operational consequences, not by mathematical extensions that conceal their singular behavior.^{1; 5}

We have presented a thought experiment involving an inelastic rope and a falling clock, showing that no horizon crossing ever occurs. The rope can be extended indefinitely, the clock continues to flash, and the observer receives each signal in perfect one-second intervals forever. The clock does not disappear. It does not freeze. It remains accessible, observable, and recoverable at any time. No part of the system becomes causally disconnected—not now, not in the future, and not in principle.

What fails is not the rope, but the assumption that an event horizon is physically crossed. The proper length required to reach the horizon diverges, even though the falling clock experiences only finite proper time. This contradiction reveals the limits of using proper time alone to define physical evolution. The crossing never becomes real because no process confirms it, and no observer witnesses it. It is a fiction of extended coordinates, not a feature of the causal structure.

We have argued that coordinate systems which allow interior evolution commit the fungibility fallacy, treating time and space as interchangeable. This mathematical maneuver creates an interior domain with no operational meaning. It replaces causal sequence with geometric extension, and assigns reality to events that cannot be confirmed in any frame that preserves continuity.^{3; 4}

In rejecting this interpretation, we revive a more physically grounded alternative. The collapsing star never completes its fall. It forms a “frozen” object with a surface just above the Schwarzschild radius, indistinguishable from a black hole in all classical observations, yet without an interior. Its mass remains suspended at the edge of time, potentially subject to slow quantum evaporation,^c.

This view restores causality to the heart of black hole physics. It replaces

^cFrom the perspective of infalling matter, any quantum-based evaporation process would determine its actual fate: not horizon crossing, but disintegration just above it.^{11; 12; 13}

imaginary interiors with measurable exteriors. It discards mathematical idealization in favor of physical verification. And it honors Einstein's enduring insight: that physics is not about what can be written, but about what can be observed. The horizon, in this interpretation, marks not a boundary to be crossed, but a limit imposed by causality itself. It is the point where clocks stall, signals fail, and physical passage becomes impossible. It is not an entrance to a deeper structure beyond—it is an end.

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